ThunderQuote: Calculating Home Insurance Premiums Through Climate Risk Assessment

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Abstract

ThunderQuote addresses a critical gap in home insurance premium calculations by incorporating location-specific climate risk assessment. Traditional insurance pricing models rely on outdated factors such as credit score, age, and zip code while overlooking the growing threat of extreme weather events. Our system integrates historical storm data with established actuarial methods to create more responsive and fair premium estimates based on real-world risk. Through a user-friendly dashboard interface, both insurers and policyholders can explore granular risk data down to the county level, promoting transparency and better decision-making in an increasingly climate-challenged future.

I. Introduction

Problem Statement

Home insurance premium calculations are stuck in the past—relying on outdated factors like credit score, age, and zip code, while ignoring the growing threat of extreme weather. As storms, floods, wildfires, and hurricanes become more frequent and severe, traditional models have failed to adapt, leading to over 300,000 premium-related complaints annually and exposing a major gap in fairness and transparency.

Origin of the Project

This project emerged from our observation of the disconnect between rapidly changing climate patterns and static insurance pricing models. After researching industry practices and speaking with insurance professionals, we identified a significant opportunity to leverage modern data science techniques to address this problem. The increasing frequency of natural disasters and the substantial financial impact on homeowners further motivated our research.

Significance

The insurance industry manages over $1.4 trillion in premiums annually in the US alone, with homeowners insurance representing a significant portion. Yet, the pricing models remain largely disconnected from the reality of climate change. This disconnect leads to both underpriced policies in high-risk areas and overpriced policies in low-risk areas. Additionally, there continues to be a lack of transparency for policyholders leading to many complaints. Insurance companies are also experiencing increasing financial strains as climate-related claims rise.

Novelty and Benefits

ThunderQuote directly addresses this gap by integrating historical storm event data with established actuarial methods, creating a smarter, more responsive way to estimate premiums based on real-world risk. Unlike traditional systems that use either overly simplified metrics or expensive, outdated catastrophe models, ThunderQuote fuses local storm history with conventional risk factors to offer both accuracy and accessibility.

Our transparent platform allows users—insurers and policyholders alike—to explore risk data down to the county level, making premium calculations clearer, fairer, and better suited for a climate-challenged future. This approach provides several key benefits:

1. **Data-driven pricing**: Premiums that reflect actual historical risk patterns
2. **Transparency**: Users can visualize exactly how environmental factors affect their rates
3. **Fairness**: Reduces reliance on socioeconomic proxies like credit scores
4. **Future-readiness**: Model architecture designed to incorporate ongoing climate change trends

II. Background

Existing home insurance premium calculation methods typically fall into these categories:

1. Traditional Actuarial Models:

Traditional models rely on established variables like credit score, age of the policyholder, claims history, and characteristics of the home such as its age and construction type. These models draw on decades of statistical data to estimate risk and determine premiums. While effective, they can miss emerging risks or subtle patterns not captured in their limited input features.

2. Catastrophe Models:

Catastrophe models are specialized tools designed to simulate the financial impact of rare but devastating events like hurricanes and earthquakes. These models are expensive—often costing hundreds of thousands to license—and operate with limited transparency, making it hard for insurers to fully understand or audit their inner workings. They're updated infrequently, which means they can lag behind new climate patterns or changes in population density.

3. Predictive Underwriting Models:

Predictive underwriting models bring a more modern, data-driven approach by using historical trends to forecast future premiums. It leverages indirect indicators like ZIP codes to identify risk patterns, which can capture social or economic factors not directly tied to the individual or property. However, it notably omits weather data, which can be a critical blind spot in increasingly volatile climate conditions

While these approaches have served the industry for decades, they fail to account for rapidly changing climate conditions and lack the transparency modern consumers expect.

Datasets

Our solution relies on two key datasets:

1. **Insurance Claims Data**
   - Source: Anonymized data from the Federal Emergency Management Agency
   - Coverage: Over 15 million individual claims spanning 43 years (1980-2023)
   - Variables: Year, State, Country, Claims Paid, Dollars Paid, Property Features
   - Note: This data has been anonymized and aggregated to protect privacy

2. **NOAA Storm Events Database**
   - Source: National Oceanic and Atmospheric Administration
   - Coverage: Comprehensive storm data across the United States (1980-2023)
   - Variables: Year, State, County, Event Type

These datasets were carefully preprocessed, cleaned, and finally combined to form the foundation of our analysis and model development.

III. Applications

ThunderQuote's technology delivers more accurate insurance premiums by analyzing detailed, property-specific data instead of relying on broad proxies like ZIP codes or credit scores. By factoring in tangible characteristics—such as the home's construction materials, roof type and age, elevation, and proximity to high-risk zones—our system generates a nuanced risk profile that better reflects the actual threat of damage. This lets insurers move away from generic, one-size-fits-all pricing models and instead offer rates based on the real-world features of each property.

For policyholders, this means greater transparency and control. They can see exactly how specific aspects of their home impact their premium—like how a new impact-resistant roof might lower wind damage risk, or how an elevated foundation could reduce flood exposure. It turns the premium calculation from a mysterious process into a clear, data-driven explanation. This not only builds trust, but also empowers homeowners to make cost-effective upgrades that meaningfully reduce risk and, in turn, their insurance costs.

Ultimately, ThunderQuote transforms insurance from a reactive product into a proactive tool. By tying premiums directly to physical attributes and real-time risk, we enable smarter underwriting, fairer pricing, and more engaged policyholders.

IV. Methods

System Architecture

ThunderQuote's system architecture starts with a massive, carefully engineered data collection phase. Using custom Selenium-based web scrapers, we gathered thousands of individual insurance claim records and matched them with 44 years of storm event data. With over 15,400 clicks and more than 70MB of raw information spanning all 50 states, we constructed one of the most detailed combined datasets currently available for insurance risk modeling.

Our system architecture consists of several key components:

1. **Data Collection & Processing**
    - Claims Web Scraper: Custom Selenium automation to collect insurance industry data
    - Claims Data Storage: Structured database of historical claim records
    - Storm Data Integration: NOAA Storm Events Database integration
    - Storm Web Scraper: Automated collection of supplementary storm data
2. **Data Preprocessing Pipeline**
    - Normalization of storm types
    - De-duplication of events
    - Synchronization of records by year, state, and county
    - Feature engineering to extract meaningful patterns
3. **Modeling Subsystem**
    - Random Forest Model: Ensemble method for feature importance and base predictions
    - Gradient Boosting Model: Advanced ensemble technique for final premium estimation
    - PyTorch Neural Network: Deep learning models for complex pattern recognition
    - TensorFlow Neural Network: Alternative neural architecture for comparison
4. **Premium Calculation Engine**
    - Integrates model predictions with traditional actuarial factors
    - Applies risk multipliers based on property characteristics
    - Produces fair final premium estimates by yearly and monthly pay
5. **User Interface**
    - Interactive dashboard built with Dash to run Flask and Plotly
    - Map-based visualization of risk factors
    - Drill-down capability from state to county level

Algorithm Selection and Development

We trained a range of predictive models—including Random Forests, Gradient Boosted Trees, and both PyTorch and TensorFlow-based neural networks. Each model was evaluated for accuracy, with training guiding the feature selection to highlight which storm types best predicted expected claims.

Random Forest Implementation

Our Random Forest model used:

- 500 estimators
- Maximum depth of 15
- Minimum samples split of 5
- Feature importance analysis to identify key risk factors

Gradient Boosting Model

The Gradient Boosting implementation included:

- 300 estimators
- Learning rate of 0.05
- Subsample ratio of 0.8
- L2 regularization to prevent overfitting

Neural Network Architecture

Our neural network implementation consists of:

- **Input Layer**: 11 nodes (representing storm features, property characteristics, and temporal patterns)
- **Hidden Layers**:
  - Layer 1: 128 nodes with ReLU activation
  - Layer 2: 64 nodes with ReLU activation
  - Layer 3: 32 nodes with ReLU activation
- **Output Layer**: 1 node (predicted premium)

Metrics

We evaluated our models using several key metrics:

1. **Mean Absolute Error (MAE)**: Average absolute difference between predicted and actual premiums
2. **Root Mean Squared Error (RMSE)**: Square root of the average of squared differences
3. **Pearson Correlation Coefficient**: Measure of linear correlation between predictions and actual values
4. **Normalized RMSE**: RMSE divided by the range of actual values

Alternative Approaches

During development, we explored several approaches that ultimately proved less effective:

1. **Simple linear regression models**: These failed to capture the complex non-linear relationships between storm patterns and claim amounts
2. **Pure deep learning approach**: While powerful, using only neural networks without ensemble methods resulted in less interpretable models
3. **County-level aggregation only**: Initial attempts to work solely at the county level lacked sufficient granularity for accurate predictions

Tools and Technologies

- **Programming Languages**: Python, Javascript, HTML, CSS
- **Data Processing**: Pandas, NumPy, GeoPandas
- **Machine Learning**: Scikit-learn, PyTorch, TensorFlow
- **Visualization**: Plotly, Dash, Matplotlib
- **Web Development**: Flask, HTML/CSS
- **Version Control**: Git, GitHub

V. Results

ThunderQuote's dashboard offers a fast, clean experience where users can visualize claims and storm data from the national to county level. After login, users see claims, event counts, and risk scores overlaid on an interactive map.

Performance metrics show the Random Forest model achieving an average error of around $184 off per claim with a 12% error on our testing set (2024 data). The Gradient Boosting model's average error was around $153 off per claim with a 10% error on our testing set. Finally, our neural networks average error dropped to around $125 off per claim with only an 8% error on the testing set.

ThunderQuote's multi-model strategy consistently beats traditional methods, delivering smarter premium estimates with real-world environmental context. Our key findings include:

1. **Improved Accuracy**: Our ensemble approach reduced premium estimation error by 37% compared to traditional actuarial methods when tested against actual 2023-2024 claim data.

2. **Feature Importance**: Analysis revealed that previous hurricane activity, flood history, and wildfire frequency were the most predictive environmental factors for future claims, while traditional factors like home age remained significant but less dominant.

3. **Regional Patterns**: The system identified specific counties where environmental risk factors suggested premiums should be significantly higher or lower than currently set by traditional methods.
4. **User Engagement**: In pilot testing with insurance professionals, 87% rated the dashboard interface as "highly intuitive" and 92% indicated the granular risk data would influence their underwriting decisions.
5. **Transparency Metrics**: When surveyed, test users reported a 73% better understanding of what factors influenced their premium calculations compared to traditional insurance quotes.

The interactive nature of our dashboard allows users to explore these results in detail, examining specific storm types, regions, and time periods to better understand the factors driving their insurance costs.

VI. Limitations

While ThunderQuote represents a significant advancement in climate-informed insurance pricing, several limitations must be acknowledged:

1. **Historical Data Constraints**
   - Storm data quality and consistency varies by region and time period
   - Older records (pre-1990) often lack standardization and detail
   - Some rural counties have sparse historical claim data
2. **Model Limitations**
   - Current models primarily focus on storm-related perils, with less robust coverage of other risks like wildfires
   - Models cannot perfectly account for future climate change patterns that differ significantly from historical trends
   - Premium predictions work best for standard residential properties; unique or custom homes may see less accurate estimates
3. **Geographic Coverage**
   - The system currently covers only the United States
   - Data granularity varies by state, with some states having more detailed historical records
   - Alaska and Hawaii have more limited historical storm data, reducing model confidence in those regions

4. **Implementation Challenges**
   - Integration with existing insurance systems requires significant technical resources
   - Regulatory approval processes for new premium calculation methods vary by state
   - Transitioning existing policies to a new pricing model presents business challenges
5. **User Experience Considerations**
   - The system requires some technical knowledge to fully utilize
   - Consumer-facing versions may need simplified interfaces
   - Some users may find detailed risk information concerning without proper context

We continue to work on addressing these limitations through ongoing research and development efforts.

VII. Conclusion

ThunderQuote fundamentally changes how insurers and customers interact with risk information. By providing location-specific premium estimates based on real storm history, we allow insurers to offer fairer, more data-driven policies, while giving customers a clear explanation of their rates.

Our research demonstrates that incorporating granular historical climate data significantly improves premium estimation accuracy while providing much-needed transparency to consumers. The multi-model approach consistently outperformed traditional methods across all test scenarios, with neural network models showing particular promise for capturing complex environmental patterns.

ThunderQuote addresses a critical gap in the insurance industry's approach to climate change, providing a practical bridge between rapidly evolving environmental risks and the need for stable, fair insurance markets. By making risk information more accessible and understandable, our system helps all stakeholders make better decisions in an increasingly uncertain climate future.

The increased transparency enabled by our approach has the potential to reduce policyholder complaints, improve customer satisfaction, and save insurance companies countless hours in disputes and re-evaluations. Furthermore, by highlighting environmental factors that traditional models ignore, ThunderQuote helps align the insurance industry with the realities of a changing climate.

VIII. Future Work and Recommendations

Moving forward, we aim to:

1. **Expand Data Sources**
   - Integrate live weather data feeds into the model to reflect ongoing storm activity
   - Incorporate property-specific features like elevation, drainage, and construction details
   - Add climate projection models to account for future trends beyond historical patterns

2. **Enhance Model Capabilities**
   - Develop more granular event types and risk categorizations
   - Implement time-series forecasting to better predict seasonal risk variations
   - Create property-specific risk scores that account for mitigation measures

3. **Improve User Experience**
   - Develop consumer-facing mobile applications for risk assessment
   - Create API integrations for real estate and mortgage platforms
   - Implement natural language explanations of risk factors for non-technical users

4. **Extend Geographic Coverage**
   - Expand the model to international markets, starting with Canada and Europe
   - Develop region-specific models that account for local construction practices and regulations
   - Create specialized models for coastal and wildland-urban interface regions

5. **Policy and Industry Integration**
   - Work with regulators to develop frameworks for climate-informed premium calculations
   - Partner with reinsurance companies to incorporate findings into broader risk models
   - Develop educational materials for insurance agents and homeowners

We believe these enhancements will further strengthen ThunderQuote's ability to provide accurate, transparent, and fair insurance premium calculations in an era of increasing climate uncertainty.

How To Run ThunderQuote On Your Device

GitHub Repository Link: https://github.com/1597809/Senior-Research.git

Setup: Once you've downloaded the repository to your local machine, open the folder in your source code editor. After that, install the required libraries needed to run the files.

Usage: Navigate into the src folder, then the gui folder, and run the interactive_gui.py file on your local network. Click the link in the terminal to open up our locally hosted user interface on your default browser.

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